

Dockerfile

A **Dockerfile** is a script-like file used to define the blueprint for creating a Docker image. It contains a series of instructions and configurations that tell Docker how to build the image. Once built, the image can be used to create lightweight, portable containers.

Key Features of a Dockerfile:

1. **Base Image:** Specifies the starting point for the image, such as a Linux distribution or a language runtime.
2. **Commands:** Includes instructions to copy files, install dependencies, configure settings, expose ports, and define the entry point for the container.
3. **Layered Architecture:** Each instruction in the Dockerfile creates a new layer, enabling efficient storage and updates.

Customized Docker Image

A **customized Docker image** is an image that you create yourself, based on your specific application requirements. Unlike official images (which are pre-built by Docker or community maintainers), a customized image is tailored to include all the dependencies, configurations, libraries, and tools your application needs to run in a containerized environment.

Customized Docker images are typically built from a **Dockerfile**, which contains instructions for setting up the environment and application. The primary goal is to create a container that can run your specific application seamlessly, with all the necessary components included.

Dockerfile Attributes

FROM: Specifies the base image to start building the new image.

WORKDIR: Sets the working directory for any subsequent RUN, CMD, or ENTRYPOINT instructions.

COPY: Copies files or directories from the host machine to the image.

ADD: Similar to COPY, but with added functionality (e.g., can retrieve files from a URL or unpack tar archives).

RUN: Executes a command during the image build process (e.g., to install dependencies or configure software).

CMD: Defines the default command to run when the container starts. Can be overridden at runtime

ENTRYPOINT: Configures a container to run as an executable. It works similarly to CMD but is more rigid and usually not overridden.

EXPOSE: Declares the port on which the container will listen at runtime.

ENV: Sets environment variables in the image.

ARG: Defines build-time variables that can be passed to the Docker build process.

VOLUME: Creates a mount point to allow persistent data storage.

LABEL: Adds metadata to the image.

USER: Sets the user for executing subsequent commands.

Launched ec2 instance

EC2

S3

CodeDeploy

CodePipeline

RDS

VPC

IAM

CloudFormation

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EC2 Global View

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Instances

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Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Find Instance by attribute or tag (case-sensitive)

All states ▼

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☒	Name 🔗 ▼	Instance ID	Instance state ▼	Instance type ▼	Status check	Alarm status	Availability Zone ▼	Public IPv4 DI
☒	Docker	i-006b633af99c3c44f	Running 🔍	t2.micro	Initializing	View alarms +	us-east-1a	ec2-98-80-118-128.compute-1.amazonaws.com

i-006b633af99c3c44f (Docker)

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Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary Info

Instance ID

i-006b633af99c3c44f

IPv6 address

—

Hostname type

IP name: ip-172-31-27-162.ec2.internal

Public IPv4 address

98.80.118.128 | [open address 🔗](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-27-162.ec2.internal

Private IPv4 addresses

172.31.27.162

Public IPv4 DNS

ec2-98-80-118-128.compute-1.amazonaws.com | [open address 🔗](#)

Logged in as a root user by sudo -i

```
[root@ip-172-31-27-162 ~]# docker --version
Docker version 25.0.6, build 32b99dd
[root@ip-172-31-27-162 ~]# systemctl enable docker
Created symlink from /etc/systemd/system/multi-user.target.wants/docker.service to /usr/lib/systemd/system/docker.service.
[root@ip-172-31-27-162 ~]# systemctl start docker
[root@ip-172-31-27-162 ~]# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; vendor preset: disabled)
   Active: active (running) since Thu 2024-12-26 12:59:37 UTC; 15s ago
     Docs: https://docs.docker.com
   Process: 3529 ExecStartPre=/usr/libexec/docker/docker-setup-runtimes.sh (code=exited, status=0/SUCCESS)
   Process: 3528 ExecStartPre=/bin/mkdir -p /run/docker (code=exited, status=0/SUCCESS)
 Main PID: 3532 (dockerd)
    Tasks: 7
   Memory: 28.1M
   CGroup: /system.slice/docker.service
           └─3532 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock --default-ulimit nofile=32768:65536

Dec 26 12:59:36 ip-172-31-27-162.ec2.internal systemd[1]: Starting Docker Application Container Engine...
Dec 26 12:59:36 ip-172-31-27-162.ec2.internal dockerd[3532]: time="2024-12-26T12:59:36.880436452Z" level=info msg="Starting up"
Dec 26 12:59:36 ip-172-31-27-162.ec2.internal dockerd[3532]: time="2024-12-26T12:59:36.925618130Z" level=info msg="Loading cont...art."
Dec 26 12:59:37 ip-172-31-27-162.ec2.internal dockerd[3532]: time="2024-12-26T12:59:37.130790193Z" level=info msg="Loading cont...one."
Dec 26 12:59:37 ip-172-31-27-162.ec2.internal dockerd[3532]: time="2024-12-26T12:59:37.143784506Z" level=info msg="Docker daemo...5.0.6
Dec 26 12:59:37 ip-172-31-27-162.ec2.internal dockerd[3532]: time="2024-12-26T12:59:37.144127477Z" level=info msg="Daemon has c...tion"
Dec 26 12:59:37 ip-172-31-27-162.ec2.internal dockerd[3532]: time="2024-12-26T12:59:37.181194553Z" level=info msg="API listen o...sock"
Dec 26 12:59:37 ip-172-31-27-162.ec2.internal systemd[1]: Started Docker Application Container Engine.
Hint: Some lines were ellipsized, use -l to show in full.
[root@ip-172-31-27-162 ~]#
```


Opening 8081 port in security group



Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info				
sgr-08e79e49b27fda709	SSH ▼	TCP	22	Cu... ▼			0.0.0.0/0 ✕		Delete
-	Custom TCP ▼	TCP	8081	An... ▼	0.0.0.0/0		0.0.0.0/0 ✕		Delete

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Accessed nginx



⚠ Not secure 98.80.118.128:8081

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Created index.html file

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Homepage</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 0;
      padding: 20px;
      background-color: #f9f9f9;
    }
    h1 {
      color: #333;
    }
    p {
      color: #555;
    }
    img {
      max-width: 100%;
      height: auto;
      border-radius: 8px;
    }
  </style>
</head>
<body>
  <h1>Welcome to My Homepage</h1>
  <p>This is my first web page with an image!</p>
  
  <p>Thank you for visiting!</p>
</body>
</html>
```

index.html" 34L, 882B

Adding COPY and VOLUME commands in Dockerfile

```
FROM nginx:1.20
LABEL Name="kechiya"
RUN apt update -y && apt install git -y && apt install wget -y
EXPOSE 80
CMD ["/usr/sbin/nginx", "-g", "daemon off;"]
COPY index.html /usr/share/nginx/html/
VOLUME ["/usr/share/nginx/html"]
```

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[root@ip-172-31-27-162 ~]# vi index.html
[root@ip-172-31-27-162 ~]# vi Dockerfile
[root@ip-172-31-27-162 ~]# docker build -t kechiya/mydockerfile .
[+] Building 0.3s (9/9) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 267B                             0.0s
=> [internal] load metadata for docker.io/library/nginx:1.20    0.1s
=> [auth] library/nginx:pull token for registry-1.docker.io     0.0s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                   0.0s
=> [1/3] FROM docker.io/library/nginx:1.20@sha256:38f8cld9613f3f42e7969c3b1dd5c3277e635d4576713e6453c6193e66270a6d 0.0s
=> CACHED [2/3] RUN apt update -y && apt install git -y && apt install wget -y 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 921B                                 0.0s
=> [3/3] COPY index.html /usr/share/nginx/html/                0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                          0.0s
=> => writing image sha256:0c6d31a0800e53077ae39ed94a98f62800ffe91367f0e660fc8b42daa024cacb 0.0s
=> => naming to docker.io/kechiya/mydockerfile                 0.0s
[root@ip-172-31-27-162 ~]# docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
kechiya/mydockerfile latest       0c6d31a0800e     41 seconds ago  245MB
<none>              <none>      81eeb56141df     About an hour ago 245MB
[root@ip-172-31-27-162 ~]# docker run -it --name kechiya -p 8081:80 0c6d31a0800e
docker: Error response from daemon: Conflict. The container name "/kechiya" is already in use by container "3d3a5fbd4e29e0d59de2b52e55826dabaf8e32338d24278165dc8508c9255301". You have to remove (or rename) that container to be able to reuse that name.
See 'docker run --help'.
[root@ip-172-31-27-162 ~]# docker run -it --name kechiya1 -p 8081:80 0c6d31a0800e
2024/12/26 15:04:26 [notice] 1#1: using the "epoll" event method
2024/12/26 15:04:26 [notice] 1#1: nginx/1.20.2
2024/12/26 15:04:26 [notice] 1#1: built by gcc 10.2.1 20210110 (Debian 10.2.1-6)
2024/12/26 15:04:26 [notice] 1#1: OS: Linux 5.10.230-223.885.amzn2.x86_64
2024/12/26 15:04:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 32768:65536
2024/12/26 15:04:26 [notice] 1#1: start worker processes
2024/12/26 15:04:26 [notice] 1#1: start worker process 7

```

Accessed customized image



Welcome to My Homepage

This is my first web page with an image!

